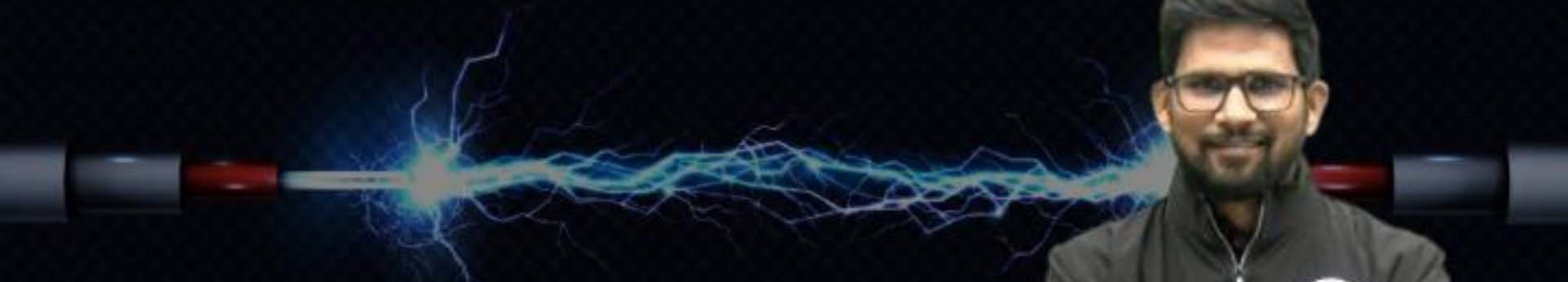


COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 11

Combinational Circuit



By- Chandan Gupta Sir



Recap of Previous Lecture

MUX

DeMUX



Topics to be Covered

Decoder

Encoder

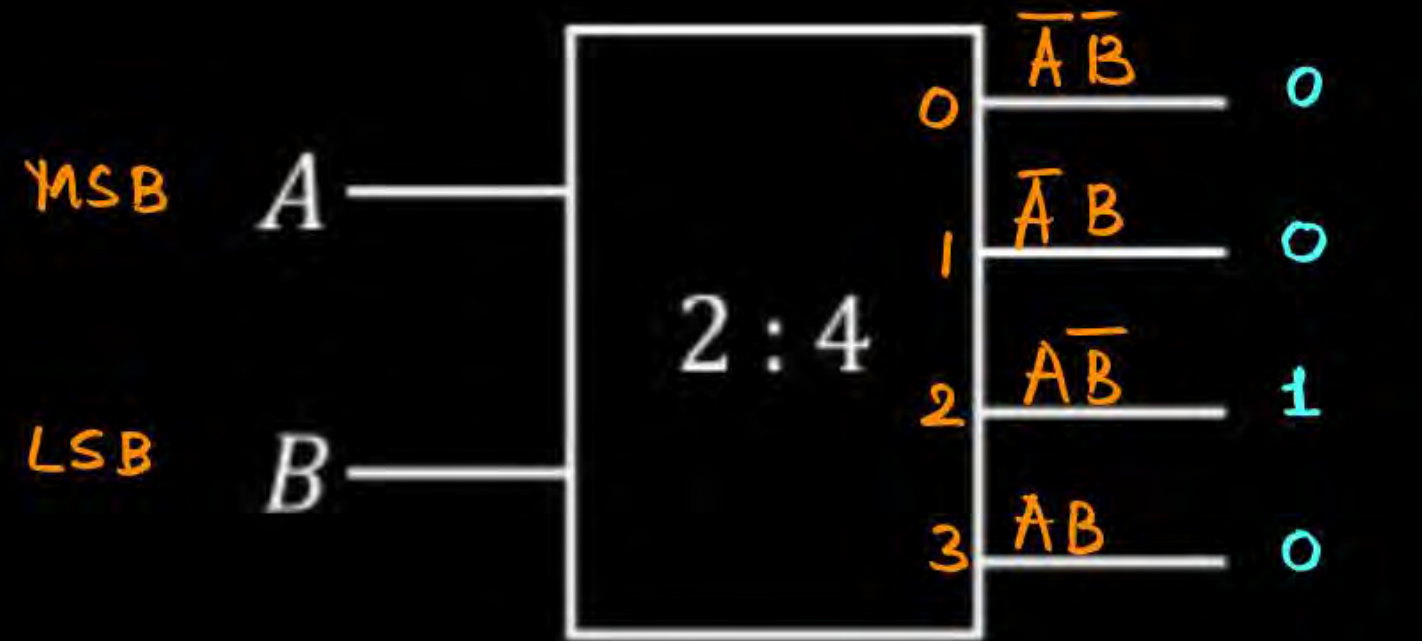
[Decoder]



- What is Decoder?

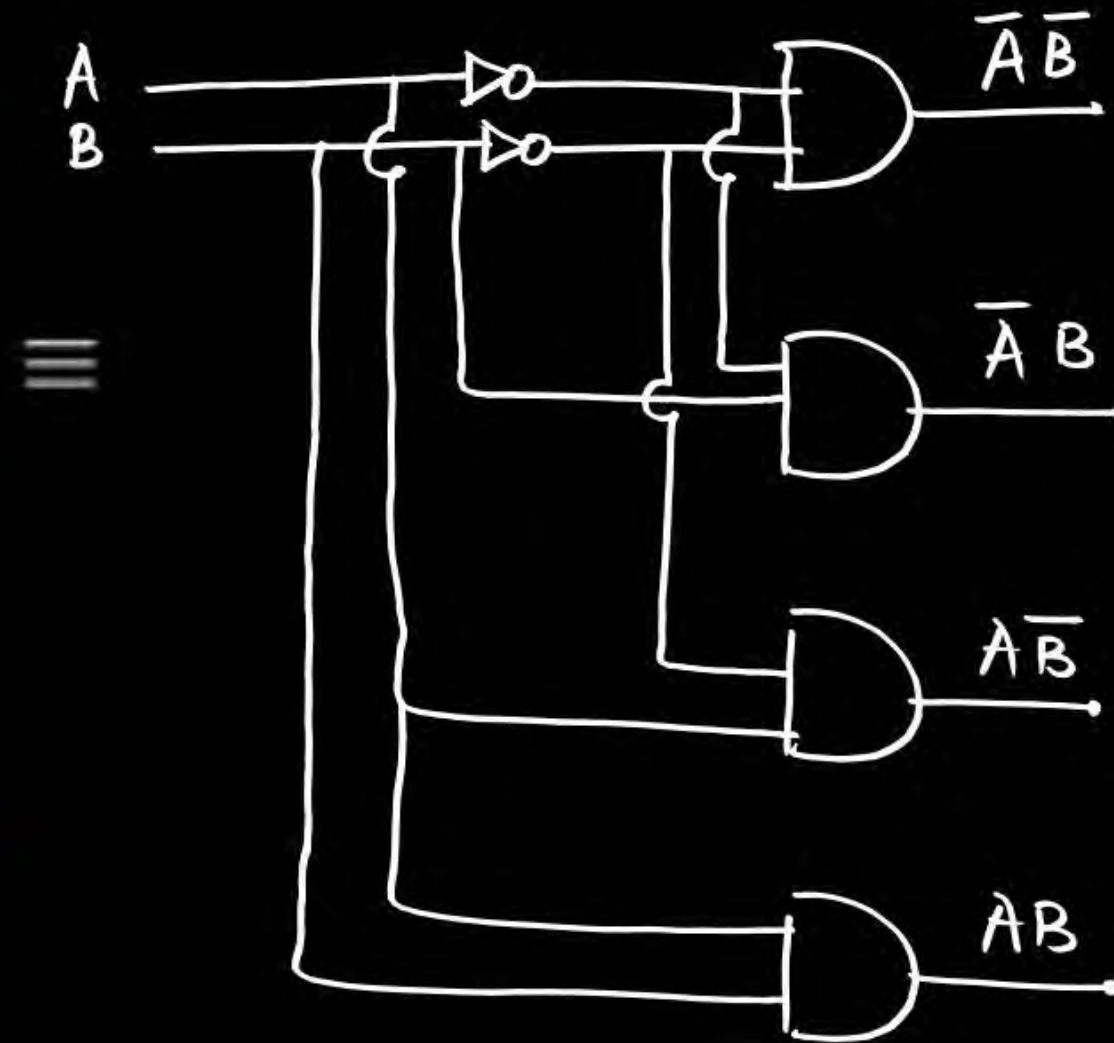
- It is a comb. ckt having many i/p lines and many o/p lines.
- Order: $n:2^n$
- It is basically used to convert binary into other codes.
 - eg. 3:8 decoder $\rightarrow$ binary to octal
 - 4:16 decoder $\rightarrow$ binary to hexadecimal

- 2 : 4 Decoder

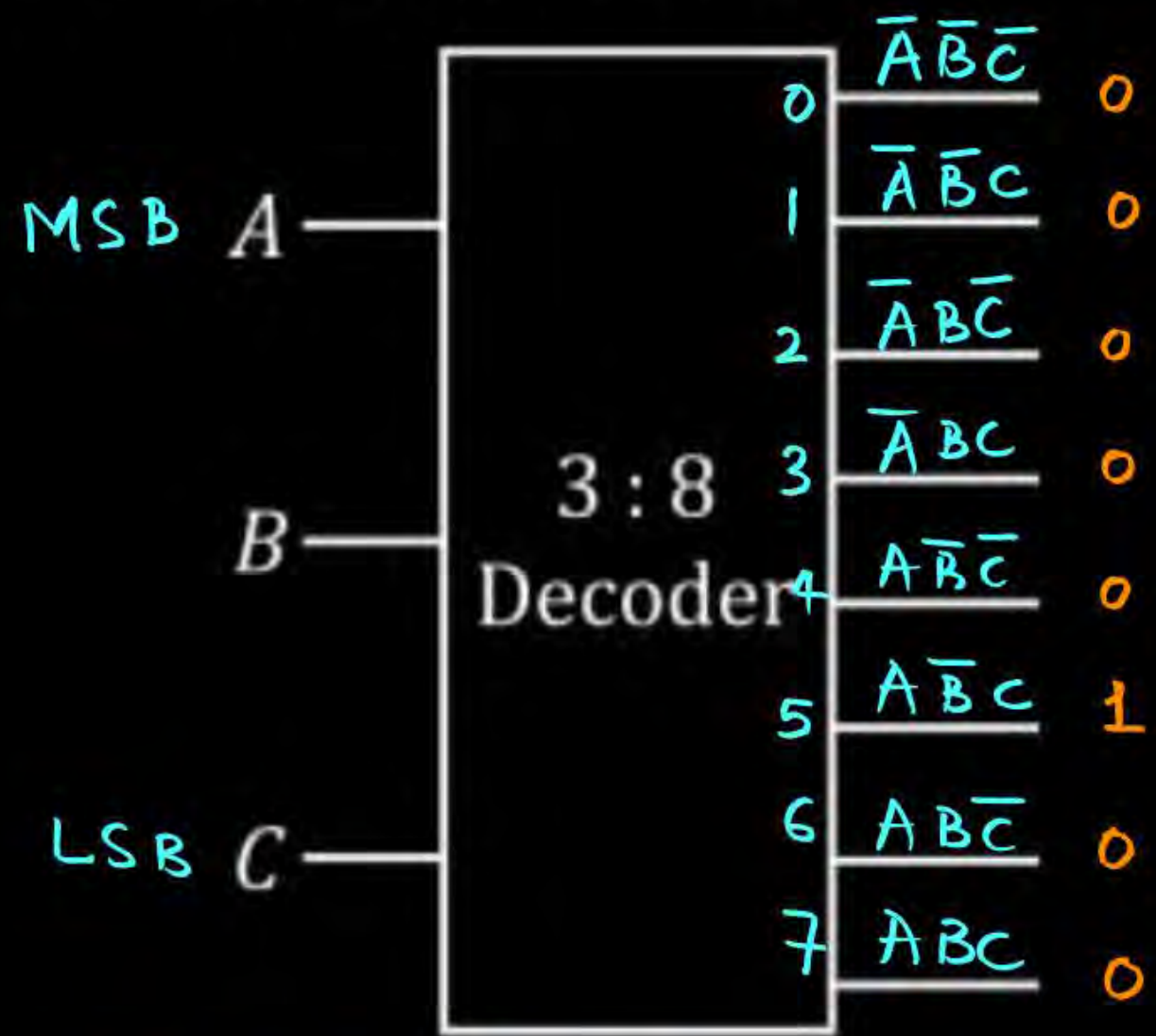


$A=1 \& B=0 \rightarrow P_{in}=1$

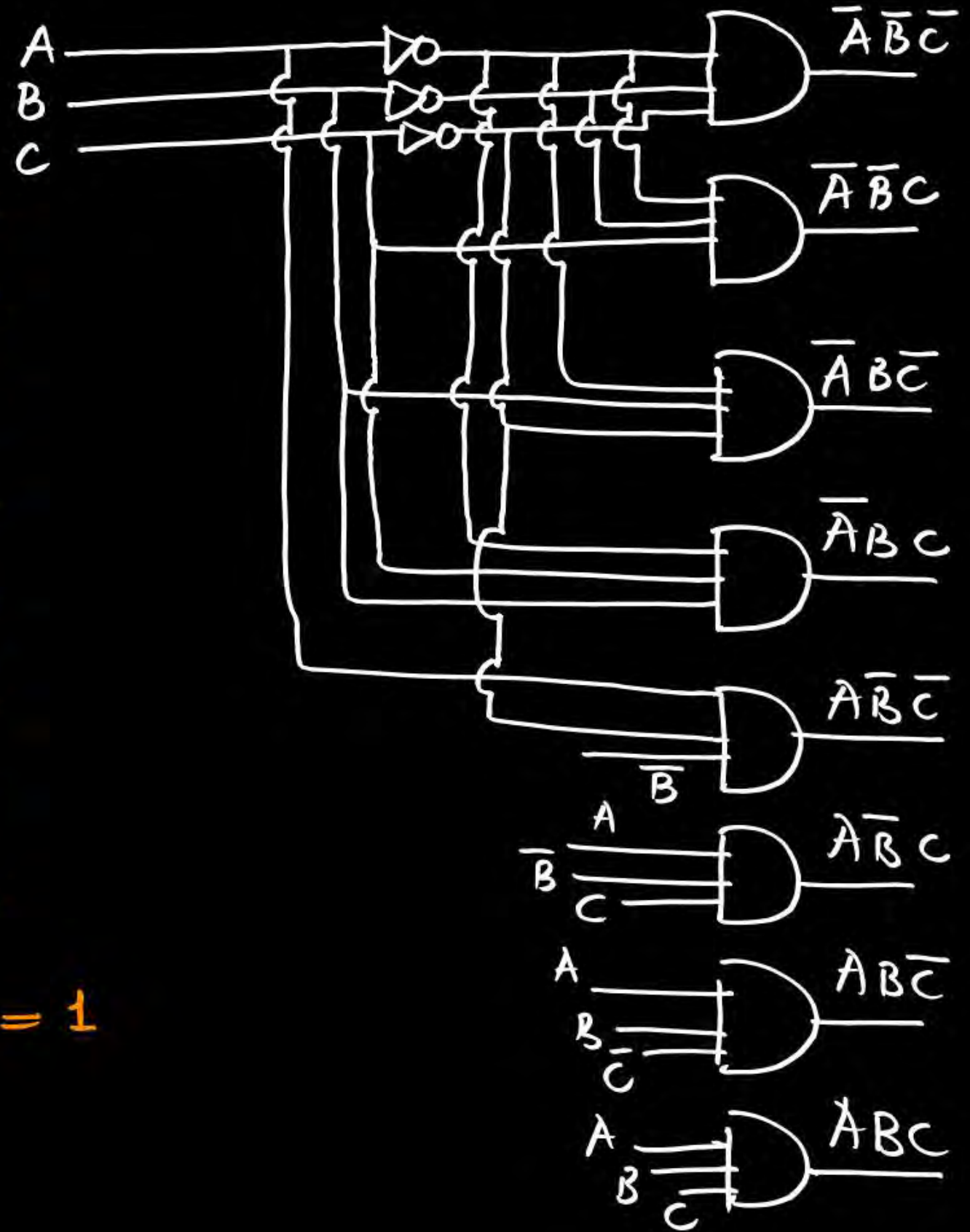
Internal Circuit



• 3 : 8 Decoder

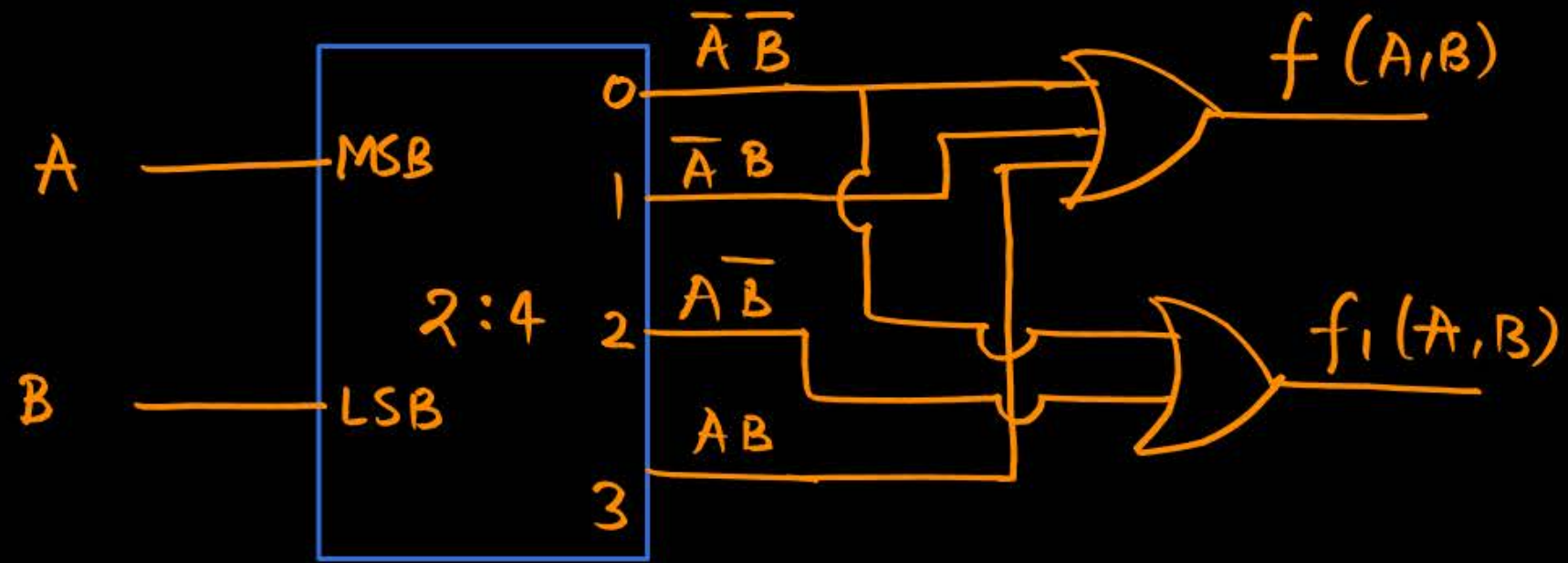


$A=1, \& B=0 \& C=1 \longrightarrow \text{Pin 5} = 1$



- $f(A,B) = \Sigma(0,1,3) = \bar{A}\bar{B} + \bar{A}B + AB$

$$f_1(A,B) = \Sigma(0,2) = \bar{A}\bar{B} + A\bar{B}$$



[Full adder implementation using Decoder + OR gate]

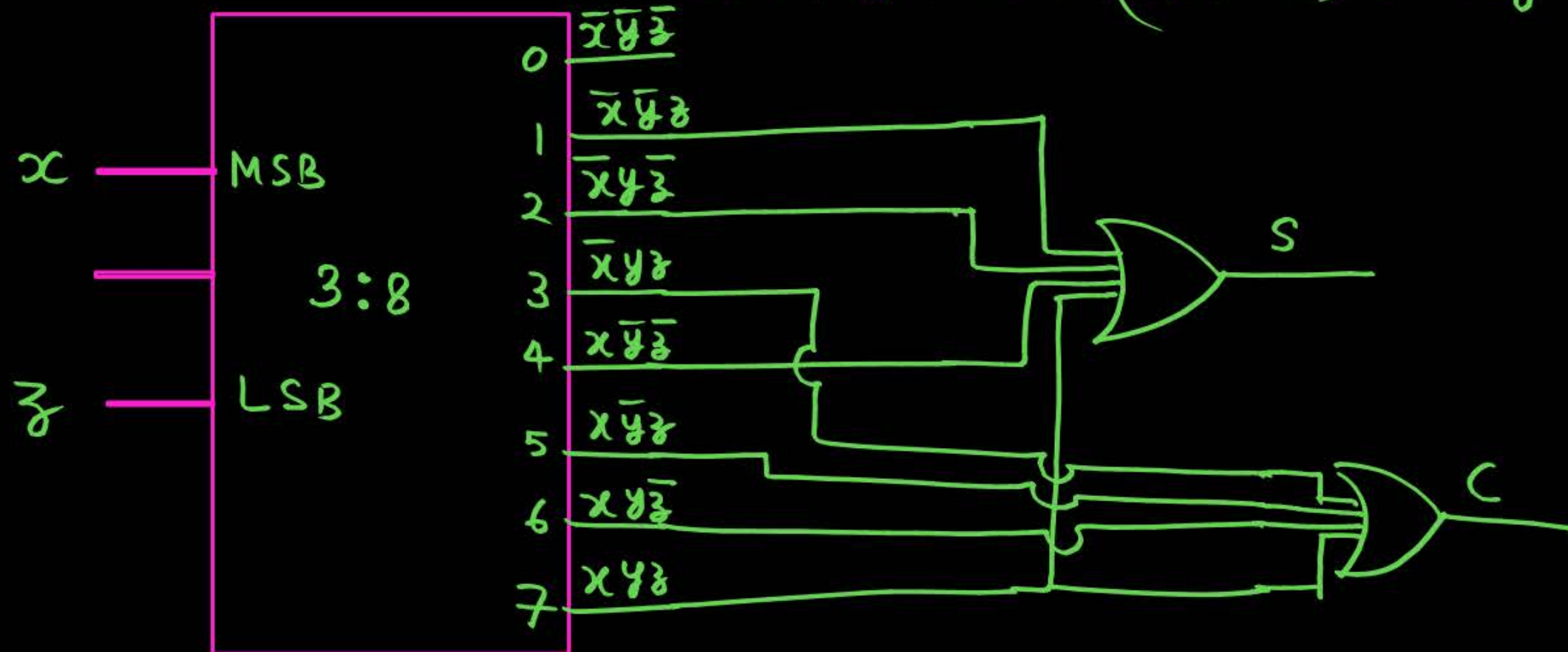


- Full adder using 3 : 8 decoder + OR gate :

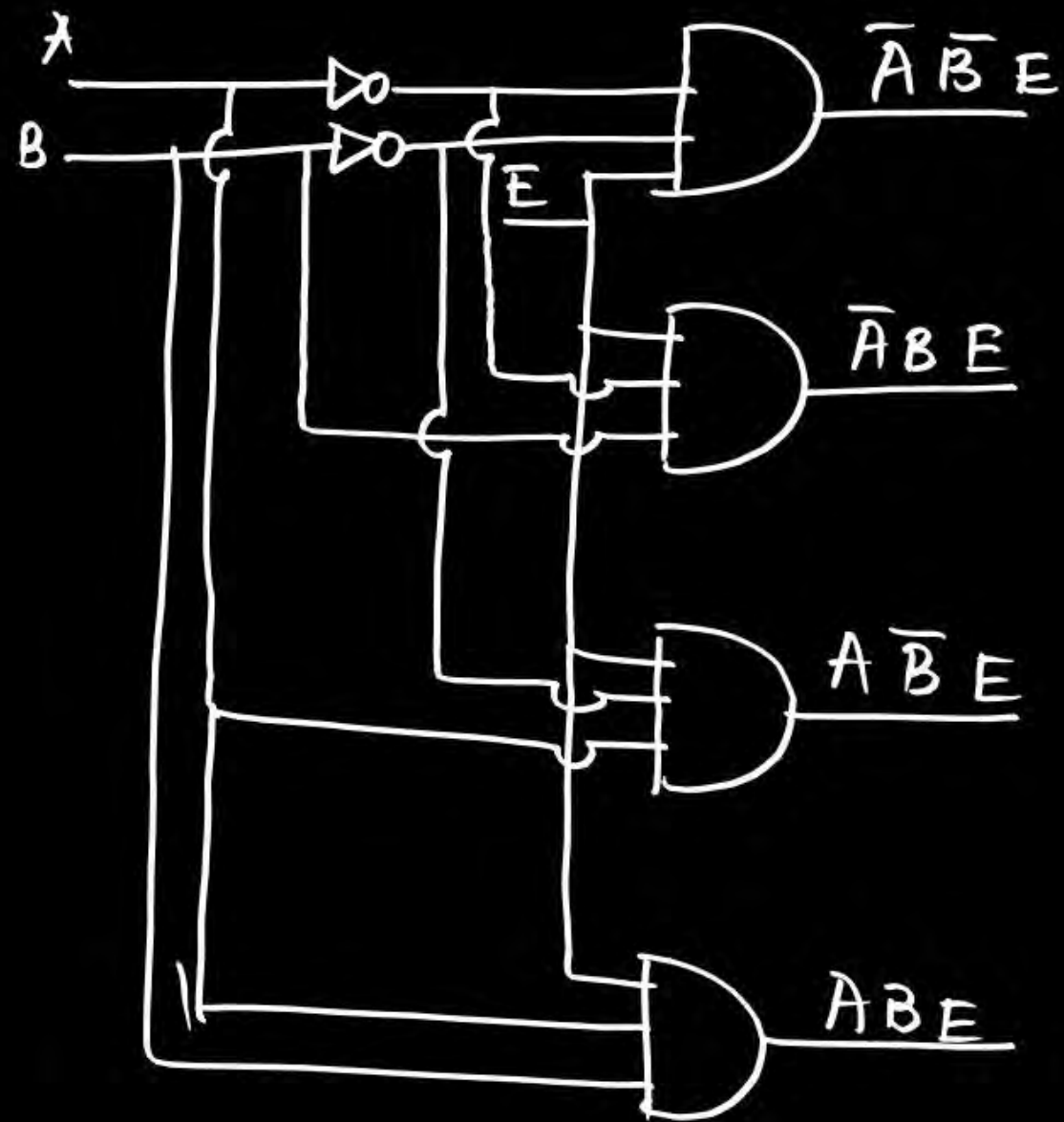
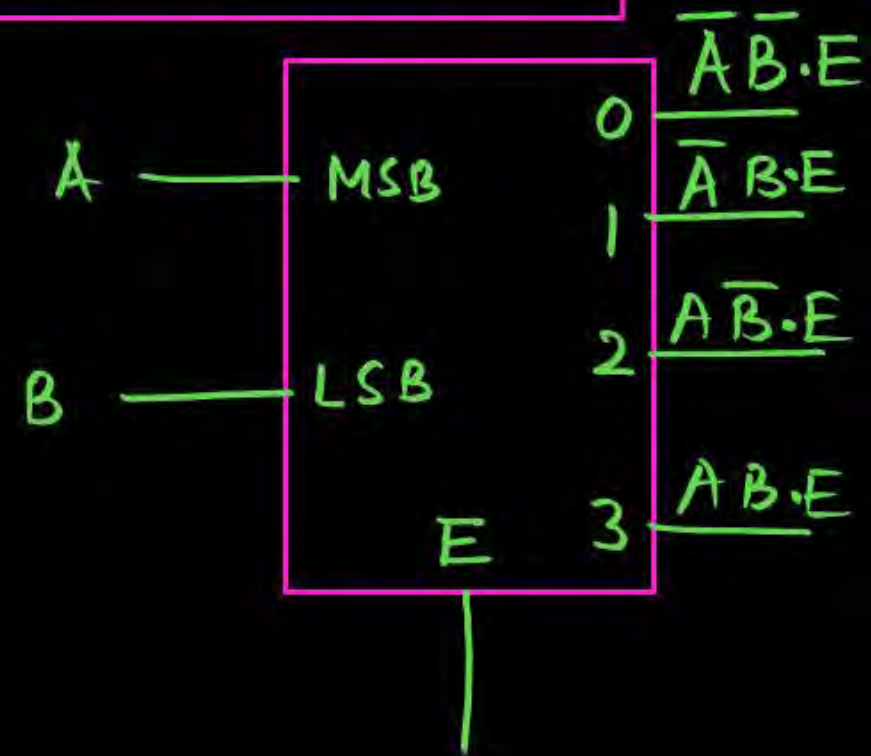
3-i/p lines (x, y, z)

$$S(x, y, z) = \Sigma(1, 2, 4, 7)$$

$$C(x, y, z) = \Sigma(3, 5, 6, 7) = xy + yz + zx$$



Decoder with enable i/p



[Comparison of DeMUX and Decoder circuit]

- How their internal circuits are same ?

Decoder with enable i/p has same internal circuitry as of same order DeMUX.

A	MSB	0	$(A+B)$	1
		1	$(A+\bar{B})$	1
B	LSB	2	$(\bar{A}+B)$	0
		3	$(\bar{A}+\bar{B})$	1

$A=1, B=0$

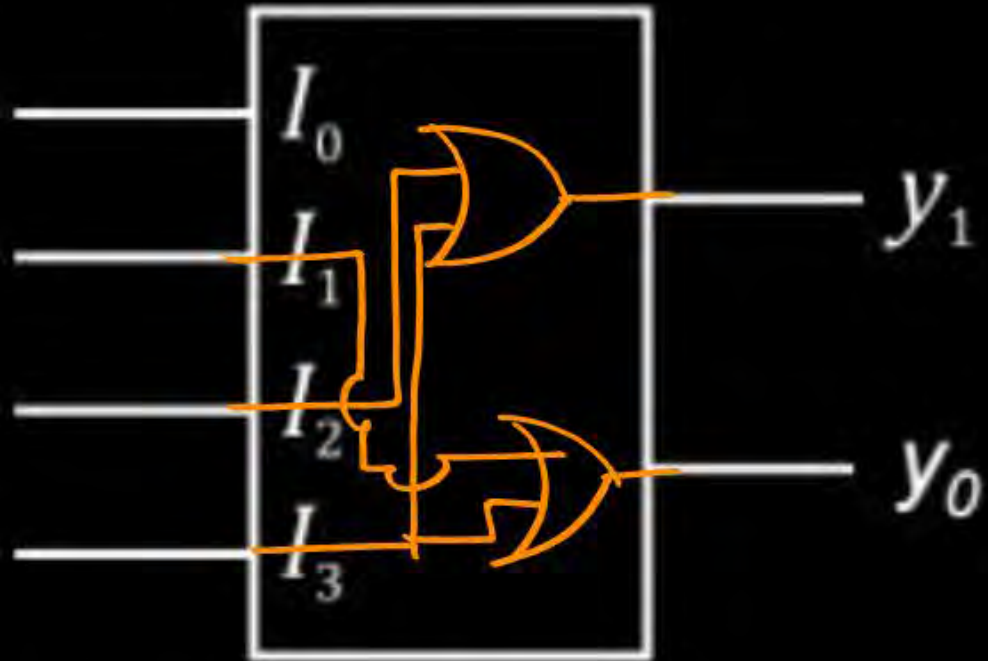
[Encoder]



- What is encoder?

- It is comb. CKt having many i/p and many o/p's.
- $2^n : n$
- It is basically used to convert other codes into binary.
 - 8:3 Encoder → Octal to binary
 - 16:4 Encoder → Hexadecimal to binary.

- 4 : 2 Encoder :



$$y_1(I_0, I_1, I_2, I_3) = I_2 + I_3$$

$$y_0(I_0, I_1, I_2, I_3) = I_1 + I_3$$

Truth Table :

	I_0	I_1	I_2	I_3	y_1	y_0
8	1	0	0	0	0	0
4	0	1	0	0	0	1
2	0	0	1	0	1	0
1	0	0	0	1	1	1

$$y_1[I_0, I_1, I_2, I_3] = \Sigma(1, 2) + d\Sigma(0, 3, 5, 6, 7, 9 - - 15)$$

$$y_0[I_0, I_1, I_2, I_3] = \Sigma(1, 4) + d\Sigma(0, 3, 5, 6, 7, 9 - - 15)$$

y_1

	$\overline{I_2}\overline{I_3}$	$\overline{I_2}I_3$	$I_2\overline{I_3}$	I_2I_3
$\overline{I_0}\overline{I_1}$	X	1	X	1
$\overline{I_0}I_1$		X	X	X
$I_0\overline{I_1}$	X	X	X	X
I_0I_1		X	X	X

$\downarrow I_3$ $\downarrow I_2$

$$y_1 = (I_2 + I_3)$$

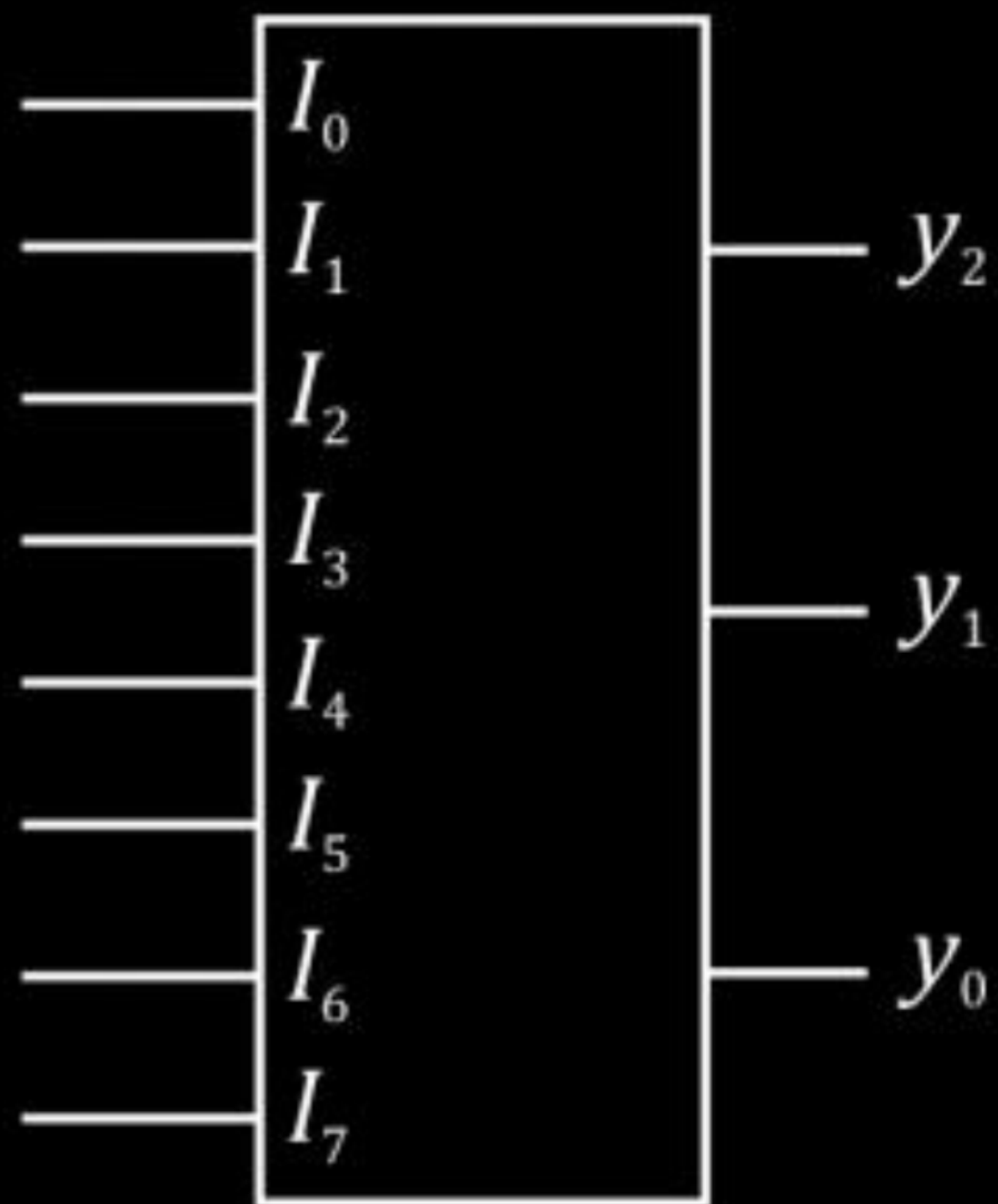
y_0

	$\overline{I_2}\overline{I_3}$	$\overline{I_2}I_3$	$I_2\overline{I_3}$	I_2I_3
$\overline{I_0}\overline{I_1}$	X	1	X	
$\overline{I_0}I_1$	1	X	X	X
$I_0\overline{I_1}$	X	X	X	X
I_0I_1		X	X	X

$\downarrow I_3$

$$y_0 = (I_1 + I_3)$$

- 8 : 3 Encoder :



Truth Table :

I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7		y_2	y_1	y_0
1	0	0	0	0	0	0	0		0	0	0
0	1	0	0	0	0	0	0		0	0	1
0	0	1	0	0	0	0	0		0	1	0
0	0	0	1	0	0	0	0		0	1	1
0	0	0	0	1	0	0	0		1	0	0
0	0	0	0	0	1	0	0		1	0	1
0	0	0	0	0	0	1	0		1	1	0
0	0	0	0	0	0	0	1		1	1	1

248- don't care conditions



$$y_2 = I_4 + I_5 + I_6 + I_7$$

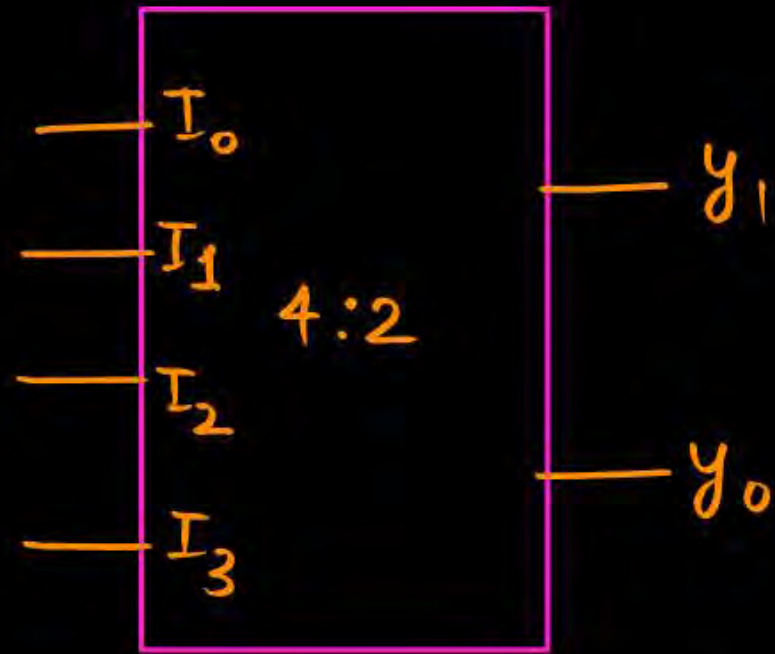
$$y_1 = I_2 + I_3 + I_6 + I_7$$

$$y_0 = I_1 + I_3 + I_5 + I_7$$

$$\begin{matrix} 2^2 & 2^1 & 2^0 \\ y_2 & y_1 & y_0 \end{matrix}$$

$$\begin{aligned} y_2 &= (I_4 + I_5 + I_6 + I_7) \\ y_1 &= (I_2 + I_3 + I_6 + I_7) \\ y_0 &= (I_1 + I_3 + I_5 + I_7) \end{aligned}$$

Priority Encoder



2024

Priority $I_3 > I_2 > I_1 > I_0$

$I_3 \rightarrow 11$

$I_2 \rightarrow 10$

$I_1 \rightarrow 01$

$I_0 \rightarrow 00$

	I_3	I_2	I_1	I_0	y_1	y_0
0 ✓	0	0	0	0	X	X
1	0	0	0	1	0	0
2	0	0	1	0	0	1
3	0	0	1	1	0	1
4	0	1	0	0	1	0
5	0	1	0	1	1	0
6	0	1	1	0	1	0
7	0	1	1	1	1	0
8	1	0	0	0	1	1
9	1	0	0	1	1	1
10	1	0	1	0	1	1
11	1	0	1	1	1	1
12	1	1	0	0	1	1
13	1	1	0	1	1	1
14	1	1	1	0	1	1
15	1	1	1	1	1	1



$$y_1[I_3, I_2, I_1, I_0] = \sum (4 - 15) + d \leq 0$$

$$y_0[I_3, I_2, I_1, I_0] = \sum (2, 3, 8 - 15) + d \leq 0$$



2 Minute Summary



→ Decoder
→ Encoder.

Thank you

GW
Soldiers !

